SAFETY DATA SHEET

Date: May 5, 2015

Section 1	**Identification of Product and Supplier** Product Name: Glo SB-1 Manufacturer: Globe Metallurgical, Inc. P.O. Box County Road 32 Beverly, OH 45715 Tel: 740-984-2361 Product Code: N/A Emergency Telephone No.: Chemtrec 1-800-424-9300 Product Use: Alloy additive Restrictions: None identified Chemical Formula: Varies Appearance: Grey metallic color Odor: None
Section 2	**Hazards Identification** <u>GHS Classification</u> **Health**: Airborne dust generated through the use or handling of the product may result in respiratory tract and/or eye discomfort. Product that has come in contact with water, acids or bases may form toxic gases (see section 10). **Physical**: Material is not known to be a water reactive flammable solid material based on United Nations recognized analytical methods. Dust from this product under certain conditions may be combustible in a dust cloud form. **This product does not create combustible dust when shipped. It is considered not to be used in downstream processes that would create dust.** OSHA Hazard Category: Not classified GHS Category: None <u>GHS Pictogram Label(s)</u> None

Signal Word
None

Hazard Statements
No known significant effects or critical hazards.

Precautionary Statements
*Keep away from contact with water, because of possible formation of flammable and noxious gases.
*Avoid breathing dust, as it may irritate and dehydrate mucous membranes.
*Keep away from oxidizing agents and halogen acids.
*Airborne dust under certain conditions may be combustible.
*Avoid skin contact – Dust may irritate skin.
*Avoid eye contact – Dust may irritate and lead to dryness of the eyes.
*Do not ingest.

Section 3 **Composition/Information on Ingredients**

Ingredient	CAS No#	Percent %
Silicon	7440-21-3	70-80
Iron	7439-89-6	18-28
Calcium	7440-70-2	0.4-1.7
Chromium	7440-47-3	0.0-0.05
Aluminum	7429-90-5	0.4-1.7
Barium	7440-39-3	0.4-1.7

Section 5 **Firefighting Measures**

Dry ferrosilicon in the form of lumps or granules (Bottom Screen > 200 Mesh [0.074 mm]) is not combustible.
Extinguishing Media: Dry powder, dry sand, or CO2
Special Information: Exposing Ferrosilicon dust to water may result in the formation of noxious gases. Ferrosilicon airborne dust may under certain conditions cause dust explosions. Wear appropriate Personal Protective Equipment (See Section 8).

Section 6 **Accidental Release Measures**

Use collection methods that minimize creation of airborne dust. High efficiency vacuuming is recommended to recover spilled dry dust. Wet product must be kept away from dry, and must not be stored in sealed containers.

Section 7 **Handling and Storage**

Handling: Prevent the creation of airborne dust. Avoid ignition sources. Wear appropriate personal protective equipment. Keep material dry. Do not re-seal any wet materials. Wet product added to molten metal may cause explosions. Inert atmosphere recommended when sizing to minus 20 mesh or lower.

Storage: Store in well ventilated dry areas. Store away from water, acids and bases.

Section 8 **Exposure Controls and Personal Protection**

Airborne Exposure Limits:
Based on an 8 Hour TWA:

Component	*OSHA PEL*	*ACGIH TLV*
Silicon	TWA 15 mg/m3 (total) TWA 5 mg/m3 (resp)	TWA 10 mg/m3 (total)
Chromium	TWA 0.5 mg/m3 **see NIOSH Guide Appendix C*	TWA 0.05 mg/m3
Aluminum	TWA 15 mg/m3 (total) TWA 5 mg/m3 (resp)	TWA 10 mg/m3 Total
Barium	TWA 0.5 mg/m3	TWA 0.5 mg/m3

Ventilation System:
Local exhaust suggested in any processing areas. Ensure good ventilation.

Personal Respirators (NIOSH Approved):
Use an appropriate filtering respirator when any airborne contaminates reach or exceed the Permissible Exposure Limit (PEL). In poorly ventilated areas, such as confined spaces, a self-contained breathing apparatus or a clean air supplied respirator should be worn.

Skin Protection:
Avoid any contact with skin. Wear protective gloves.

Eye Protection:
Avoid any contact with eyes. Wear appropriate eye protection, and have access to eye wash stations.

Other Control Measures:
Do not breathe respirable dusts.

Section 9 **Physical and Chemical Properties**

Appearance: Grey metallic solid.	Melting Point: 1200 – 1250 ˚C
Odor: None	Freezing Point: Not available
Solubility: Insoluble in water	Flash Point: Not available
Specific Gravity: 2 – 4 (H20 = 1.0)	Flammability (solid): No ignition
pH: Not available	Vapor Density (Air = 1): Not available
% Volatiles by Volume @ 21C (70F): Not available	Vapor Pressure (mm Hg): Not available
Boiling Point: Not Available	Evaporation Rate (BuAc = 1): Not available

Section 10 **Stability and Reactivity**

Stability: Stable under normal temperatures and dry conditions.

Conditions to Avoid: Avoid sparks and hot work (e.g. welding and grinding etc.) in areas with high concentrations of airborne dust.

Incompatibilities: Moisture, strong oxidizing agents, and halogen acids. Reacts

	rapidly with hydrofluoric and nitric acid, as well as, molten alkali. Hazardous Decomposition Products: The toxic gases phosphine and arsine may be formed if product gets in contact with moisture, acids, or bases. The flammable gas may form if product gets in contact with water. Toxic gases may form if product is reacted with Hydrofluoric acid or Nitric acid. Hazardous Polymerization: Has not been observed.
Section 11	**Toxicological Information** Material is not known to be toxic. However, if product gets in contact with moisture, acids or bases the toxic gases phosphine and arsine may be formed. Phosphine and arsine may be absorbed from dust being deposited on mucous membranes.
Section 12	**Ecological Information** Environmental Fate: Not currently known Eco-toxicological Fate: Not currently known
Section 13	**Disposal Considerations** Recovered material should be recycled, if possible. Ensure the material is disposed of in accordance with all Federal, State and Local regulations.

Section 14 **Transport Information**

DOT 49 CFR	IATA Air	IMDG Vessel	ADR
Not Regulated for Transportation	Not Regulated for Transportation	Not Regulated for Transportation	Not Regulated for Transportation

***Ferrosilicon minus 200 mesh with the composition as described in section 3 has been tested in accordance with the Code of Federal Regulations, Title 49, and the United Nations Transport of Dangerous Goods – Manual of Tests and Criteria, Fifth Edition (2009), and has passed the test. Therefore, ferrosilicon is not classified as a Division 4.3 dangerous when wet product.**

Section 15 **Other Regulatory Information**

TSCA Listed: YES

Section 16 **Other Information**

NFPA Ratings *estimated*	HMIS Ratings:
Health - 1	Health - 1
Flammability - 0	Flammability - 0
Reactivity - 1	Physical Hazard - 1

All information, recommendations, and suggestions, appearing herein concerning our products are based on data believed to be accurate and reliable. Since the actual use by others is beyond our control, it is the user's responsibility to determine the suitability of the product for its use and to adopt such safety precautions as may be necessary. Since the conditions of use are not under our control, Globe disclaims all liability with respect to the use of any material supplied by Globe.